

SQL Commands

DDL · DML · TCL

Core categories of SQL statements, the commands inside each, and query examples for every command.



DDL

Definition



DML

Manipulation



TCL

Transaction

Three Categories of SQL Commands

Every SQL statement falls into one of these functional groups.



DDL

Data Definition Language

Defines and modifies the structure of database objects — tables, schemas, indexes.

COMMANDS

- **CREATE**
- **ALTER**
- **DROP**
- **TRUNCATE**
- **RENAME**



DML

Data Manipulation Language

Works with the data stored inside tables — reading, adding, changing, removing rows.

COMMANDS

- **SELECT**
- **INSERT**
- **UPDATE**
- **DELETE**



TCL

Transaction Control Language

Manages transactions so a group of DML changes can be made permanent or undone.

COMMANDS

- **COMMIT**
- **ROLLBACK**
- **SAVEPOINT**
- **SET TRANSACTION**



DDL — Data Definition Language

Commands that define or change the structure of database objects

CREATE

Builds a new database object — table, view, index, schema.

ALTER

Changes the structure of an existing object (add/drop/modify columns).

DROP

Permanently deletes an object and its data from the database.

TRUNCATE

Removes all rows from a table instantly, keeping the table structure.

RENAME

Changes the name of an existing database object.

DDL — Example Queries

One sample statement for each DDL command

CREATE

```
CREATE TABLE students (  
  id INT PRIMARY KEY,  
  name VARCHAR(50),  
  age INT  
);
```

ALTER

```
ALTER TABLE students  
ADD email VARCHAR(100);  
  
-- adds a new column
```

DROP

```
DROP TABLE students;  
  
-- deletes table + all data
```

TRUNCATE / RENAME

```
TRUNCATE TABLE students;  
  
RENAME TABLE students  
TO learners;
```



DML — Data Manipulation Language

Commands that read and modify the data stored inside tables

SELECT

Retrieves rows and columns of data from one or more tables.

INSERT

Adds one or more new rows of data into a table.

UPDATE

Modifies the values of existing rows that match a condition.

DELETE

Removes rows that match a condition, one at a time (loggable, undoable).

DML — Example Queries

One sample statement for each DML command

   SELECT

```
SELECT name, age
FROM students
WHERE age > 18;
```

   INSERT

```
INSERT INTO students
(id, name, age)
VALUES (1, 'Asha', 20);
```

   UPDATE

```
UPDATE students
SET age = 21
WHERE id = 1;
```

   DELETE

```
DELETE FROM students
WHERE id = 1;
```



TCL — Transaction Control Language

Commands that manage the effect of DML changes within a transaction

COMMIT

Permanently saves all changes made in the current transaction.

ROLLBACK

Undoes changes made in the current transaction, back to the last commit or savepoint.

SAVEPOINT

Marks a point inside a transaction that you can later roll back to.

SET TRANSACTION

Sets properties for a transaction, such as its isolation level.

TCL — Example Queries

A transaction using SAVEPOINT, ROLLBACK, and COMMIT together

transaction.sql

```
UPDATE students SET age = 22 WHERE id = 1;
```

```
SAVEPOINT before_delete;
```

```
DELETE FROM students WHERE id = 2;
```

```
-- changed my mind about the delete
```

```
ROLLBACK TO before_delete;
```

```
-- keep the age update, discard nothing else
```

```
COMMIT;
```

Quick Comparison

DDL, DML, and TCL side by side

DDL

DML

TCL

Acts on

Structure	Data / rows	Transactions
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Auto-commit

Yes, always	Depends on mode	Controls it
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Can be rolled back

No (usually)	Yes	Yes — it IS rollback
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Key commands

CREATE, ALTER, DROP	SELECT, INSERT, UPDATE, DELETE	COMMIT, ROLLBACK, SAVEPOINT
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